

Answer: A

- 17** Since the simple pendulum undergoes simple harmonic motion,

$$\theta = \theta_0 \sin \omega t$$

$$0.030 = 0.050 \sin \left(\frac{2\pi}{2} t \right),$$

$$t = 0.205 \text{ s}$$

Calculating for angular speed,

$$\frac{d\theta}{dt} = \omega \theta_0 \cos \omega t$$

$$= \left(\frac{2\pi}{2} \right) (0.050) \cos \left(\frac{2\pi}{2} \times 0.205 \right)$$

$$= 0.126 \text{ rad s}^{-1}$$

Answer: D